

Sri Lanka Institute of Information Technology



Lab sheet No 05

Exercise 04

IT25100826

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IT2011 - Artificial Intelligence and Machine Learning

B.Sc. (Hons) in Information Technology

1. In this lab we will use OpenAI API to develop an application in python
2. Create an account at <https://www.deeplearning.ai>
3. Register to the free course at <https://learn.deeplearning.ai/courses/chatgpt-building-system>
4. Select the second activity

The screenshot shows the DeepLearning.AI course interface. On the left, a sidebar lists course topics: Introduction (2m), Language Models, the Chat Format and Tokens (19m), Classification (4m), Moderation (8m), Chain of Thought Reasoning (10m), and Chaining Prompts (19m). The main content area displays a video player for 'L1 Language Models, the Chat Format and Tokens' with a play button and a 'Show Transcript' button. To the right of the video is a Jupyter notebook titled 'L1 Student'. The notebook shows a code cell with the following Python code:

```
In [ ]: import os
import openai
import tiktoken
from dotenv import load_dotenv, find_dotenv
_ = load_dotenv(find_dotenv()) # read local .env file
openai.api_key = os.environ['OPENAI_API_KEY']
```

Below the code cell is a section labeled 'helper function'.

4. Execute the first code line in Jupyter notebook

```
In [ ]: import os
import openai
import tiktoken
from dotenv import load_dotenv, find_dotenv
_ = load_dotenv(find_dotenv()) # read local .env file
openai.api_key = os.environ['OPENAI_API_KEY']
```

helper function

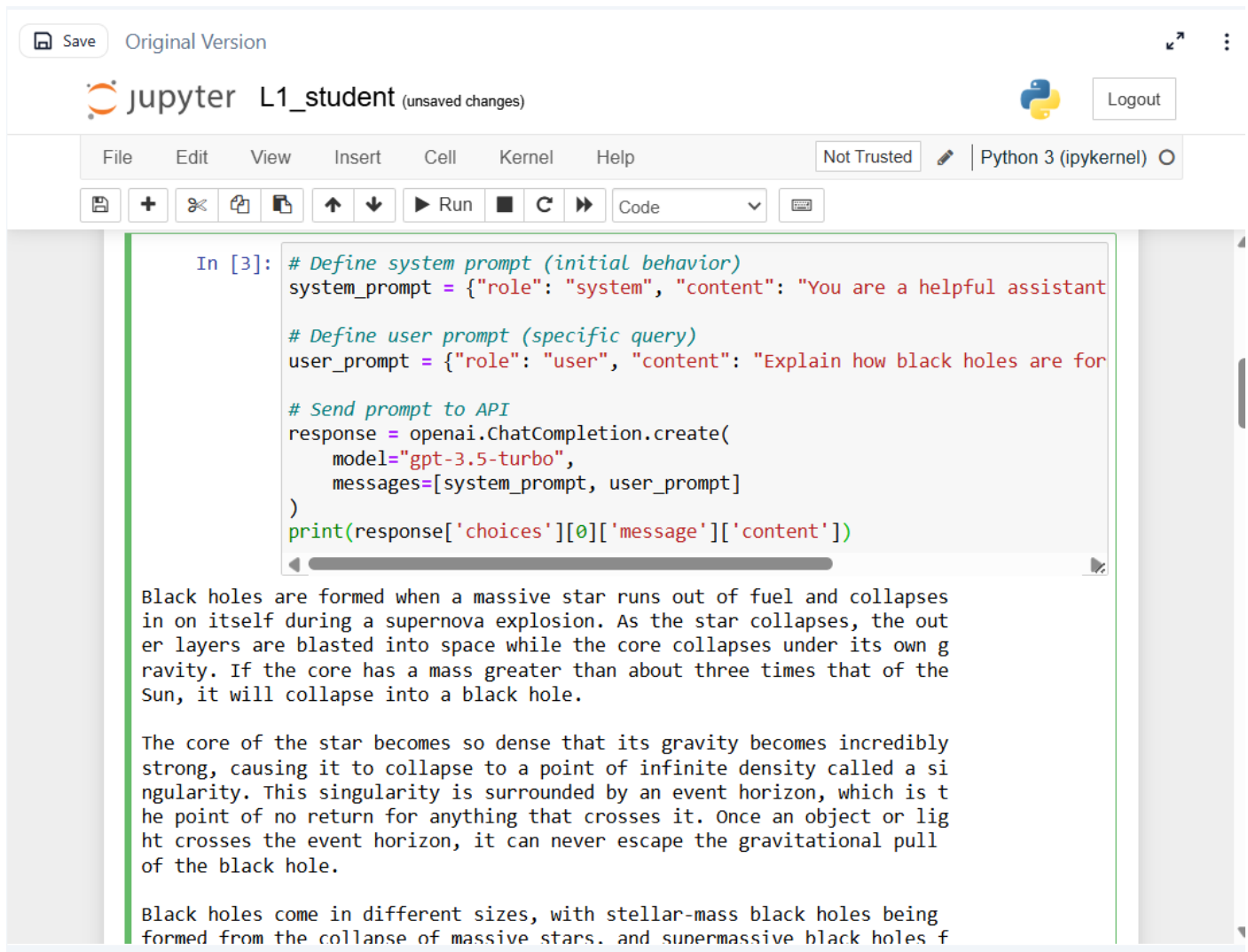
This may look familiar if you took the earlier course "ChatGPT Prompt Engineering for Developers" Course.

Throughout this course, we will use OpenAI's `gpt-3.5-turbo` model and the [chat completions endpoint](#).

This helper function will make it easier to use prompts and look at the generated outputs.

Note: In June 2023, OpenAI updated gpt-3.5-turbo. The results you see in the notebook may be slightly different than those in the video. Some of the prompts have also been slightly modified to produce the desired results.

5. Add the following code and in a codeblock and run



The screenshot shows a Jupyter Notebook interface. At the top, there's a 'Save' button and 'Original Version' text. The notebook title is 'jupyter L1_student (unsaved changes)'. The top bar includes 'File', 'Edit', 'View', 'Insert', 'Cell', 'Kernel', and 'Help' menus, along with a 'Not Trusted' warning and 'Python 3 (ipykernel)' selection. Below the menu bar is a toolbar with icons for file operations and a 'Run' button. The main area contains a code cell with the following Python code:

```
In [3]: # Define system prompt (initial behavior)
system_prompt = {"role": "system", "content": "You are a helpful assistant"}

# Define user prompt (specific query)
user_prompt = {"role": "user", "content": "Explain how black holes are formed"}

# Send prompt to API
response = openai.ChatCompletion.create(
    model="gpt-3.5-turbo",
    messages=[system_prompt, user_prompt]
)
print(response['choices'][0]['message']['content'])
```

The output of the code cell consists of two paragraphs of text:

Black holes are formed when a massive star runs out of fuel and collapses in on itself during a supernova explosion. As the star collapses, the outer layers are blasted into space while the core collapses under its own gravity. If the core has a mass greater than about three times that of the Sun, it will collapse into a black hole.

The core of the star becomes so dense that its gravity becomes incredibly strong, causing it to collapse to a point of infinite density called a singularity. This singularity is surrounded by an event horizon, which is the point of no return for anything that crosses it. Once an object or light crosses the event horizon, it can never escape the gravitational pull of the black hole.

Black holes come in different sizes, with stellar-mass black holes being formed from the collapse of massive stars, and supermassive black holes f

6. Copy the codeblock given above again to the next cell and perform the following changes
7. Give a System Prompt for a chatbot to behave as an expert event planner
8. Give a User Prompt asking for advice how to plan for an event you have to do, maybe your friends Birthday Party, or a Batch Outing

CODE

```
# Define system prompt (initial behavior)

system_prompt = {"role": "system", "content": "You are an expert event planner who helps users organize successful events with realistic budgets, schedules, activities, food, and logistics."}

# Define user prompt (specific query)

user_prompt = {"role": "user", "content": "I am planning a birthday party for my friend. There will be 15 people and my budget is Rs. 30,000. Give me a complete plan including food, decorations, activities, schedule, and estimated costs."}

# Send prompt to API

response = openai.ChatCompletion.create(
    model="gpt-3.5-turbo",
    messages=[system_prompt, user_prompt] )

print(response['choices'][0]['message']['content'])
```

ANSWER

Sure! Here's a complete plan for your friend's birthday party with a budget of Rs. 30,000 for 15 people:

Theme and Decorations:

- **Theme**: Casual Garden Party
- **Decorations**: Balloons, streamers, fairy lights, and colorful tableware
- **Estimated Cost**: Rs. 5,000

Food and Drinks:

- **Menu**:
 - Assorted finger foods (mini sandwiches, kebabs, sliders)
 - Mini pizzas
 - Cupcakes or a small birthday cake
- Soft drinks, juices, and water
- **Estimated Cost**: Rs. 10,000

Activities and Entertainment:

- **Games**:
 - DIY photo booth with props

- Scavenger hunt
- Musical chairs
- **Estimated Cost**: Rs. 2,000

Schedule:

- **2:00 PM - 2:30 PM**: Guests arrive, mingling, and welcome drinks
- **2:30 PM - 4:00 PM**: Games and activities
- **4:00 PM - 5:00 PM**: Food and cake cutting
- **5:00 PM - 6:00 PM**: Socializing, Photo booth, and thank you speech

Logistics:

- **Venue**: Your backyard or a rented garden space
- **Tables and Chairs**: Rs. 3,000 (if you need to rent)
- **Sound System**: Rs. 2,000 (for music and announcing games)

Total Estimated Cost: Rs. 20,000

With this plan, you have some buffer in your budget for any unexpected expenses or additional decorations you may want to add. Feel free to adjust the details based on your preferences and needs. Enjoy planning and hosting the birthday party!

The screenshot displays a Jupyter Notebook titled "L1_student" with the following components:

- Table of Contents (Left Sidebar):**
 - Building Systems with the ChatGPT API
 - Introduction (Video · 2m)
 - Language Models, the Chat Format and Tokens (Video with Code Example · 19m)
 - Classification (Video with Code Example · 4m)
 - Moderation (Video with Code Example · 8m)
 - Chain of Thought Reasoning (Video with Code Example · 10m)
 - Chaining Prompts (Video with Code Example · 19m)
 - Course (0%)
 - Accomplishment (0/1 assignment)
- Code Editor (Center):**

```
In [3]: # Define system prompt (initial behavior)
system_prompt = {
    "role": "system",
    "content": "You are an expert event planner who helps users organize s
}

# Define user prompt (specific query)
user_prompt = {
    "role": "user",
    "content": "I am planning a birthday party for my friend. There will b

# Send prompt to API
response = openai.ChatCompletion.create(
    model="gpt-3.5-turbo",
    messages=[system_prompt, user_prompt]
)

print(response['choices'][0]['message']['content'])
```
- Output (Bottom):**

Sure! Here's a complete plan for your friend's birthday party with a budg
et of Rs. 30,000 for 15 people:

Theme and Decorations:

 - **Theme**: Casual Garden Party
 - **Decorations**: Balloons, streamers, fairy lights, and colorful tablew
are
 - **Estimated Cost**: Rs. 5,000
- Right Sidebar:**
 - Language Models, the Ch...
 - Building Systems with the ChatGPT API (Overview, 19:30)
 - Show Transcript

```

- **Estimated Cost**: Rs. 5,000

### Food and Drinks:
- **Menu**:
  - Assorted finger foods (mini sandwiches, kebabs, sliders)
  - Mini pizzas
  - Cupcakes or a small birthday cake
  - Soft drinks, juices, and water
- **Estimated Cost**: Rs. 10,000

### Activities and Entertainment:
- **Games**:
  - DIY photo booth with props
  - Scavenger hunt
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- **Estimated Cost**: Rs. 2,000

### Schedule:
- **2:00 PM - 2:30 PM**: Guests arrive, mingling, and welcome drinks
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### Logistics:
- **Venue**: Your backyard or a rented garden space
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- **Sound System**: Rs. 2,000 (for music and announcing games)

```

```

### Total Estimated Cost: Rs. 20,000

```

With this plan, you have some buffer in your budget for any unexpected expenses or additional decorations you may want to add. Feel free to adjust the details based on your preferences and needs. Enjoy planning and hosting the birthday party!

9. Tryout a unique System prompt and User Prompt for a problem of your choice.

CODE

```

# Define system prompt (initial behavior)

```

```

system_prompt = {

```

```

    "role": "system",

```

```

    "content": "You are an expert university study coach. You help students create realistic study plans and explain difficult academic topics in simple language."

```

```

}

```

```

# Define user prompt (specific query)

```

```

user_prompt = {

```

```

    "role": "user",

```

"content": "I have exams in Probability and Statistics, Database Management Systems, and Software Engineering in two weeks. Create a daily study plan that includes revision, practice questions, and breaks."

```
}  
# Send prompt to API  
response = openai.ChatCompletion.create(  
    model="gpt-3.5-turbo",  
    messages=[system_prompt, user_prompt]  
)  
print(response['choices'][0]['message']['content'])
```

ANSWER

Sure! Here's a suggested daily study plan for your upcoming exams:

****Morning Session (8:00 AM - 12:00 PM)****

- 8:00 AM - 9:30 AM: Probability and Statistics
 - Review key concepts, formulas, and theorems
 - Solve practice problems and examples
- 9:30 AM - 9:45 AM: Short Break (15 minutes)
- 9:45 AM - 11:15 AM: Database Management Systems
 - Focus on database design, normalization, SQL queries, and transactions
 - Practice creating ER diagrams and writing SQL queries
- 11:15 AM - 12:00 PM: Software Engineering
 - Review software development life cycle models, design patterns, and testing techniques
 - Solve coding problems related to software engineering concepts

****Afternoon Session (1:00 PM - 5:00 PM)****

- 1:00 PM - 2:30 PM: Probability and Statistics
 - Work on more advanced problems and case studies
 - Review probability distributions and hypothesis testing
- 2:30 PM - 2:45 PM: Short Break (15 minutes)
- 2:45 PM - 4:15 PM: Database Management Systems

- Practice creating database schemas, normalization, and optimization techniques
- Solve sample questions on indexing, views, and transactions
- 4:15 PM - 5:00 PM: Software Engineering
 - Focus on agile methodologies, requirement engineering, and software testing
 - Review case studies on software project management

****Evening Session (6:00 PM - 8:00 PM)****

- 6:00 PM - 7:00 PM: Review the most challenging topics from each subject
- 7:00 PM - 8:00 PM: Relax and unwind, engage in a hobby or physical activity to rest your mind

Remember to take short breaks in between study sessions to rest your brain and stay hydrated. Stay focused, practice regularly, and revise consistently to ace your exams. Good luck with your study plan!

```
In [4]: # Define system prompt (initial behavior)
system_prompt = {
    "role": "system",
    "content": "You are an expert university study coach. You help student
}

# Define user prompt (specific query)
user_prompt = {
    "role": "user",
    "content": "I have exams in Probability and Statistics, Database Manag
}

# Send prompt to API
response = openai.ChatCompletion.create(
    model="gpt-3.5-turbo",
    messages=[system_prompt, user_prompt]
)

print(response['choices'][0]['message']['content'])
```

Sure! Here's a suggested daily study plan for your upcoming exams:

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 - Practice creating ER diagrams and writing SQL queries
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 - Review software development life cycle models, design patterns, and testing techniques
 - Solve coding problems related to software engineering concepts

****Afternoon Session (1:00 PM - 5:00 PM)****

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- 2:45 PM - 4:15 PM: Database Management Systems
 - Practice creating database schemas, normalization, and optimization techniques
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PROOF:

<https://learn.deeplearning.ai/courses/chatgpt-building-system/lesson/r172r/language-models%2C-the-chat-format-and-tokens>